

24. Elimination Theory and General Ideal Theory (continued)

§ 4. Arithmetical Formulation of the Concept of Dimension

A first formulation of the concept of dimension that is independent of the introduction of transformed ideals is given by

Theorem V. *The dimension of a prime ideal is given by the algebraic rank of its residue-class field. The highest dimension of an ideal is equal to the highest of the dimension numbers of its associated prime ideals.*

The first part of Theorem V was proved in the first remark to Theorem IV, where it was shown that the algebraic rank of the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$ is equal to that of the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$, hence equal to the dimension of $\mathfrak{p}$, in agreement with § 1, 8. For the proof of the second part, let $\mathfrak{m} = [\mathfrak{q}_1, \dots, \mathfrak{q}_a]$ be a shortest representation by greatest primary components. Then, first of all, no $\mathfrak{q}$ as a divisor of $\mathfrak{m}$ can possess higher dimension than $\mathfrak{m}$. But the product of all $\mathfrak{q}$ is divisible by $\mathfrak{m}$, and therefore so is the product of all elementary-divisor forms of the individual $\mathfrak{q}$'s. If, then, $n - i$ is the highest of the dimension numbers of the $\mathfrak{q}$, then $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$ and can therefore not have dimension higher than $n - i$. The dimension numbers of the $\mathfrak{q}$ agree, by Theorem I, with those of their associated prime ideals. Thus if, without introducing transformed ideals, one defines the dimension of a prime ideal by the algebraic rank of its residue-class field, then the second part of Theorem V gives the definition of the highest dimension of an arbitrary ideal.

In order, on the other hand, to grasp the concept of dimension by chains of prime ideals – again independently of the introduction of transformed ideals – one has, supplementing Theorem II, to show:

Theorem VI. *Every prime ideal different from the unit ideal possesses – after possible adjunction of indeterminates – at least one prime ideal as divisor whose dimension has decreased by exactly one unit.*

Indeed, if $\mathfrak{p}$ has dimension $(n - i) > 0$ and v denotes a new indeterminate, then

$$\mathfrak{c} = (\mathfrak{p}, x_i - v)$$

is an ideal with the required property. This follows from an isomorphic correspondence. One has $\mathfrak{c} = (\mathfrak{p}^*, x_i - v)$, where $\mathfrak{p}^*$ arises from $\mathfrak{p}$ by replacing x_i by the indeterminate v and adjoining v to the coefficient domain; the residue classes are likewise obtained by replacing the class (x_i) by (v) . But $\mathfrak{p}$ passes into itself when x_i is adjoined to $P(u)$; for from

$$h(x_i)f(x) \equiv 0 \pmod{\mathfrak{p}}, \quad h(x_i) \neq 0$$

one necessarily obtains $f(x) \equiv 0 \pmod{\mathfrak{p}}$. Otherwise $\mathfrak{p}$ would contain a polynomial depending only on x_i , and therefore also one depending only on x_n , contrary to the supposition it would have dimension at most zero. Under the assignment $v \sim x_i$, therefore, the zero class modulo $\mathfrak{c}$ necessarily corresponds to the zero class modulo $\mathfrak{p}$, and of course conversely as well. Hence there is an isomorphism between the residue-class system modulo $\mathfrak{p}$ and that modulo $\mathfrak{c}$; the latter has no zero divisors, so $\mathfrak{c}$ is a prime ideal. Under this isomorphism the algebraic dependence between (x_i) and $(P(u, x_{i+1}, \dots, x_n))$ corresponds to a relation between $(x_{i+1}), \dots, (x_n)$ with respect to $(P(u, v))$, whereas for $\lambda = 1, \dots, i - 1$ the dependences between (x_λ) and $(P(u, x_{i+1}, \dots, x_n))$ remain as such, and hence do not further lower the algebraic rank. The algebraic rank has decreased by exactly one unit. If $\mathfrak{p}$ was of dimension zero, $\mathfrak{c}$ becomes the unit ideal; the above result therefore remains valid. In the original domain $\mathfrak{S}$, consequently,

$$\bar{\mathfrak{a}} = (\bar{\mathfrak{p}}, v + v_1 y_1 + \dots + v_n y_n),$$

where v_λ is put for $-t_{i\lambda}$, is an ideal of the required kind. If, in particular, P contains infinitely many elements, then $v, v_1, \dots, v_n$ can always be specialized to elements of P in such a way that norm and elementary-divisor form, and hence the dimension of $\bar{\mathfrak{a}}$, become equal to those of the specialized ideal $\mathfrak{b}$;¹ the elementary-divisor form need not, however, remain a prime function. By Theorem V, $\mathfrak{b}$ nevertheless possesses at least one associated prime ideal of the desired dimension; passing back again to $\mathfrak{S}$, Theorem VI therefore holds there without adjoining any indeterminates.

Theorem II and Theorem VI immediately give the desired chain formulation:

Theorem VII. *A prime ideal $\mathfrak{p}$ has dimension $n - i$ if – after possible adjunction of indeterminates – there exists at least one chain of $1 + (n - i) + 1$ prime ideals*

$$\mathfrak{p}_0 = \mathfrak{p}, \quad \mathfrak{p}_1, \dots, \mathfrak{p}_{n-i}, \quad \mathfrak{p}_{n-i+1},$$

¹ Compare H.-N., the last paragraph of § 7.

each of which is a proper divisor of the immediately preceding one, whereas no such chain with a larger number of members exists. This definition also holds in the original domain, and without introducing any indeterminates, when P contains infinitely many elements.²

First it is clear that the last member of the chain must be equal to the unit ideal – which satisfies the definition of a prime ideal – since otherwise the chain could be lengthened by adjoining $\mathfrak{o}$; for $\mathfrak{o}$ itself Theorem VII gives dimension -1 , in agreement with § 1, 8. Now let $\mathfrak{p}$ be different from $\mathfrak{o}$ and of dimension $n - i$. Then the number of members of the chain can be at most $1 + (n - i) + 1$, since by Theorem II two prime ideals of the same dimension cannot occur. Conversely, by Theorem VI, after possible adjunction of new indeterminates, there exists at least one chain of this length, since the dimension has decreased by exactly one unit at every member of the chain. If one takes Theorem VII as the definition of dimension – a definition which plainly can be stated in exactly the same way in the original domain, and for which, by Theorem VI, the introduction of indeterminates becomes unnecessary when P contains infinitely many elements – then the definition of the highest dimension of an arbitrary ideal is again given by the second part of Theorem V.

§ 5. Classification of the Fundamental Ideals and of Their Components in General Ideal Theory. Zeroes of Ideals

The question is the characterization of the fundamental ideals and their components (§ 1, 7) by isolated components of the decomposition (§ 1, 13) in the sense of general ideal theory. This characterization will show the parallelism between the decomposition theorems for norms and those of general ideal theory, and will make it possible to state, for arbitrary ideals, the theory of zeroes given in § 3 for prime ideals and primary ideals. For the fundamental ideals one has

Theorem VIII. *The fundamental ideals $\mathfrak{g}_i$ of i -th stage are the isolated components of the decomposition that are uniquely determined by the totality of the associated prime ideals of dimensions $n - 1, n - 2, \dots, n - i$. If all associated prime ideals have the same dimension, then the ideal also has this dimension; that is, only one factor $R^{(i)}$ of the norm occurs. In general the norm has as many factors $R^{(i)}, R^{(i+\sigma)}, R^{(i+\tau)}, \dots$ different from unity as different dimension numbers occur among the associated prime ideals.*

Precede the proof by

Lemma VI. *If an ideal is represented as a least common multiple, $\mathfrak{m} = [\mathfrak{a}_1, \dots, \mathfrak{a}_a]$, then its fundamental ideal $\mathfrak{g}_i$ is the least common multiple of the fundamental ideals $\mathfrak{h}_i$ of i -th stage of the $\mathfrak{a}$.*

For from

$$b^{(i+1)}f \equiv 0 \pmod{\mathfrak{m}}, \quad b^{(i+1)} \neq 0$$

one obtains

$$b^{(i+1)}f \equiv 0 \pmod{\mathfrak{a}_\lambda}, \quad b^{(i+1)} \neq 0;$$

therefore $\mathfrak{g}_i$ is divisible by $[\mathfrak{h}_{i1}, \dots, \mathfrak{h}_{ia}]$. Conversely, from

$$c_\lambda^{(i+1)}f \equiv 0 \pmod{\mathfrak{a}_\lambda}, \quad c_\lambda^{(i+1)} \neq 0$$

one obtains

$$c_1^{(i+1)} \dots c_a^{(i+1)}f \equiv 0 \pmod{\mathfrak{m}}, \quad c_1^{(i+1)} \dots c_a^{(i+1)} \neq 0,$$

and hence the converse divisibility, proving the lemma.

Now let the associated prime ideals of $\mathfrak{m}$ be ordered by dimension (some dimensions may of course be absent, $j_\lambda = 0$):

$$\mathfrak{p}_{1,1}, \dots, \mathfrak{p}_{1,j_1}; \quad \mathfrak{p}_{2,1}, \dots, \mathfrak{p}_{2,j_2}; \quad \dots \quad \mathfrak{p}_{n,1}, \dots, \mathfrak{p}_{n,j_n},$$

where in general $\mathfrak{p}_{\lambda,\kappa}$ denotes a prime ideal of dimension $n - \lambda$; let $\mathfrak{q}_{\lambda,\kappa}$ be the corresponding primary ideal occurring in a fixed given decomposition of $\mathfrak{m}$. Then by definition the fundamental ideal of i -th stage is equal to the unit ideal for all $\mathfrak{q}_{\lambda,\kappa}$ for which $\lambda > i$, whereas for $\lambda \leq i$ it is, by Theorem I, equal to $\mathfrak{q}_{\lambda,\kappa}$. Therefore, by Lemma VI,

$$\mathfrak{g}_i = [\mathfrak{q}_{1,1}, \dots, \mathfrak{q}_{1,j_1}; \dots; \mathfrak{q}_{i,1}, \dots, \mathfrak{q}_{i,j_i}], \quad (5)$$

and this representation recognizes $\mathfrak{g}_i$ as an isolated component of the decomposition, since none of the prime ideals belonging to $\mathfrak{g}_i$ can be contained in one of the remaining prime ideals, all of which have lower dimension.

²This chain definition of dimension gives, in the case of an algebraic number field, dimension zero for the ordinary prime ideals, dimension -1 for the unit ideal, and dimension one for the zero ideal. Indeed the latter also satisfies the definition of prime ideals: in a field a product of non-vanishing quantities is always different from zero. Theorem II remains valid, with this definition of dimension, in algebraic number fields.

If, in particular, only associated prime ideals of dimension $n - i$ occur, then by (5) one has $\mathfrak{g}_0, \dots, \mathfrak{g}_{i-1}$ equal to the unit ideal, while $\mathfrak{g}_i = \mathfrak{g}_{i+1} = \dots = \mathfrak{m}$; hence $R_{\mathfrak{m}} = R^{(i)}$. If, however, prime ideals of different dimensions occur, say $j_i, j_{i+\sigma}, j_{i+\tau}, \dots$ are non-zero, then by (5)

$$\mathfrak{g}_0 \neq \mathfrak{g}_i \neq \mathfrak{g}_{i+\sigma} \neq \mathfrak{g}_{i+\tau} \neq \dots,$$

whereas

$$\mathfrak{g}_0 = \mathfrak{g}_1 = \dots = \mathfrak{g}_{i-1} = \mathfrak{o}, \quad \mathfrak{g}_i = \mathfrak{g}_{i+1} = \dots = \mathfrak{g}_{i+\sigma-1}, \dots;$$

therefore the factors $R^{(i)}, R^{(i+\sigma)}, R^{(i+\tau)}$, and only these, are different from unity.³

By means of Theorem VIII, Theorem I sharpens to

Theorem IX. *An ideal is primary if and only if its elementary-divisor form – and therefore its norm – is a primary function.*

That the condition is fulfilled for primary ideals was shown in Theorem I. Conversely, let the elementary-divisor form of $\mathfrak{m}$ be given by $Q^{(i)} = (P^{(i)})^\rho$, where $P^{(i)}$ denotes a prime function. By Theorem VIII all prime ideals of $\mathfrak{m}$ then have the same dimension $n - i$. From $\mathfrak{m} = [\mathfrak{q}_1, \dots, \mathfrak{q}_a]$ one obtains

$$Q^{(i)} \equiv 0 \pmod{\mathfrak{q}_\lambda}; \quad \text{hence} \quad (P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{p}_\lambda}, \quad \text{and therefore} \quad P^{(i)} \equiv 0 \pmod{\mathfrak{p}_\lambda}.$$

Since all $\mathfrak{p}_\lambda$ have dimension $n - i$ and $P^{(i)}$ is a prime function, $P^{(i)}$ becomes the elementary-divisor form of each $\mathfrak{p}_\lambda$. Thus, by the corollary to Theorem IV, all $\mathfrak{p}_\lambda$ coincide. Hence, since greatest primary components are involved, only one $\mathfrak{p}$ can occur; $\mathfrak{m}$ is primary, the special prime case for $\rho = 1$ not being excluded. Theorems VIII and IX lead to the classification of the components of the fundamental ideals as follows.

Theorem X. *The component $\mathfrak{r}_\lambda$ of the fundamental ideal $\mathfrak{g}_i$ corresponding to the greatest primary factor $Q_\lambda^{(i)} = (P_\lambda^{(i)})^{\rho_\lambda}$ of $R^{(i)}$ is given by the isolated component of the decomposition which is uniquely determined by the associated prime ideals of dimensions $n - 1, \dots, n - i + 1$ and by that prime ideal of dimension $n - i$ whose elementary-divisor form becomes equal to $P_\lambda^{(i)}$. Associated prime ideals and greatest primary factors of the norm correspond one-to-one.*

By § 1, 7 the $\mathfrak{r}_\lambda$ agree with their fundamental ideal of i -th stage and possess $\mathfrak{g}_{i-1}$ as fundamental ideal of $(i - 1)$ -st stage. Hence, by Theorem VIII respectively (5), they admit a shortest representation

$$\mathfrak{r}_\lambda = [\mathfrak{g}_{i-1}, \mathfrak{q}_{\lambda,1}, \dots, \mathfrak{q}_{\lambda,t_\lambda}],$$

where all $\mathfrak{q}$ have dimension $n - i$ and belong to distinct prime ideals. By the definition of $Q_\lambda^{(i)}$ – taking into account that $\mathfrak{r}_\lambda$ agrees with its fundamental ideal of i -th stage – one has

$$Q_\lambda^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{q}_{\lambda,\kappa}}, \quad (\kappa = 1, \dots, t_\lambda),$$

and consequently

$$(Q_\lambda^{(i)})^{\rho_\lambda} \mathfrak{r}_\lambda \equiv 0 \pmod{\mathfrak{q}_{\lambda,\kappa}}; \quad \text{hence} \quad (P_\lambda^{(i)})^{\rho_\lambda} \mathfrak{r}_\lambda \equiv 0 \pmod{\mathfrak{p}_{\lambda,\kappa}},$$

and so $P_\lambda^{(i)} \equiv 0 \pmod{\mathfrak{p}_{\lambda,\kappa}}$. As in the proof of Theorem IX it follows that in the representation of $\mathfrak{r}_\lambda$ only one $\mathfrak{q}_\lambda$ and associated $\mathfrak{p}_\lambda$ of dimension $n - i$ can occur, and that the elementary-divisor form of $\mathfrak{p}_\lambda$ is given by $P_\lambda^{(i)}$. It remains to prove that $\mathfrak{p}_\lambda$ is an associated prime ideal of $\mathfrak{m}$ and that $\mathfrak{q}_\lambda$ actually occurs in at least one representation of $\mathfrak{m}$ by greatest primary components. Then $\mathfrak{r}_\lambda$ is also recognized as the isolated component of the decomposition, since no prime ideal can be absorbed in one of equal or lower dimension, namely as the component determined by $\mathfrak{p}_\lambda$ and by all associated prime ideals of higher dimension.

For this proof it suffices to show that $\mathfrak{q}_\lambda$ occurs in a shortest representation of $\mathfrak{g}_i$, since such a representation can always, by Theorem VIII, be completed to a shortest representation of $\mathfrak{m}$. Thus it is enough to show that

$$\mathfrak{g}_i = [\mathfrak{r}_1, \dots, \mathfrak{r}_a] = [\mathfrak{g}_{i-1}, \mathfrak{q}_1, \dots, \mathfrak{q}_a]$$

becomes a shortest representation. If, for instance, $\mathfrak{q}_\lambda$ were absorbed by its complement, so that $\mathfrak{g}_{i-1} \mathfrak{q}_1 \dots \mathfrak{q}_{\lambda-1} \mathfrak{q}_{\lambda+1} \dots \mathfrak{q}_a$ were divisible by $\mathfrak{q}_\lambda$, then, because $\mathfrak{g}_{i-1} \not\equiv 0 \pmod{\mathfrak{q}_\lambda}$, it would follow that a power of some $\mathfrak{q}_\mu$ is divisible by $\mathfrak{q}_\lambda$ for $\mu \neq \lambda$. Thus the associated prime ideals $\mathfrak{p}_\mu$ and $\mathfrak{p}_\lambda$, both of the

³Taking Lemma II into account, Theorem VIII again implies that the fundamental ideals – as least common multiples of transformed ideals – become transformed ideals. Since Theorem VIII rests only on theorems from H.-N. where this fact is not used, this gives at the same time a new proof which reveals the internal reason. In H.-N. the theorem is used only in the theory of zeroes, which is newly developed here.

same dimension, would have to coincide by Theorem II; but this is impossible, since their elementary-divisor forms $P_\mu^{(i)}$ and $P_\lambda^{(i)}$ are distinct by definition. The representation of $\mathfrak{g}_i$, to which by Theorem VIII all associated prime ideals of dimension $n - i$ correspond, further shows that to every such prime ideal there really corresponds a greatest primary factor of the norm. The one-to-one correspondence is therefore proved.

Theorem X immediately gives the transfer of Theorem IV:

Theorem XI. *The individual greatest primary factors $Q_\lambda^{(i)} = (P_\lambda^{(i)})^{\rho_\lambda}$ of the norm admit a product representation*

$$Q_\lambda^{(i)} = \prod (x_i - t_{1i}\bar{y}_{1\nu} - \cdots - t_{ni}\bar{y}_{n\nu})^{\delta_\lambda \rho_\lambda} = \prod (t_{1i}(y_1 - \bar{y}_{1\nu}) + \cdots + t_{ni}(y_n - \bar{y}_{n\nu}))^{\delta_\lambda \rho_\lambda},$$

where $\bar{y}_{1\nu}, \dots, \bar{y}_{n\nu}$ represents an associated system of zeroes of the associated prime ideal $\mathfrak{p}_\lambda$, in the original indeterminates y , independent of $t_{1i}, \dots, t_{ni}$, and δ_λ has the value one or p^{f_λ} . Thus the decomposition of the norm into linear factors is completely given. The degree of $Q_\lambda^{(i)}$ gives the degree, over the residue-class subfield $(P(u, x_{i+1}, \dots, x_n))$ obtained by quotient formation, of the subring of residue classes, represented by residue classes from $\mathfrak{g}_{i-1}$, modulo the isolated component $\mathfrak{r}_\lambda$; for primary ideals this is therefore the ring of all residue classes. When new indeterminates v are adjoined, one further obtains the decomposition

$$Q_\lambda^{(i)}(u, v) = \prod (z - v_1\bar{x}_{1\nu} - \cdots - v_i\bar{x}_{i\nu})^{\delta_\lambda \rho_\lambda}.$$

Since $P_\lambda^{(i)}$ has been recognized by Theorem X as the elementary-divisor form of $\mathfrak{p}_\lambda$, this is in fact only the substitution of the decomposition of Theorem IV. The remark on the degree is only another formulation of what was said in § 1, 5 and 7.⁴

§ 6. Characterization of Prime Ideals and Proper Primary Ideals by Elementary-Divisor Form and Norm

In Theorem IX primary ideals were characterized by elementary-divisor form and norm, but only in such a way that the prime case was to be regarded as a special case of primary. We now need to separate prime ideals from proper primary ideals; the considerations will run parallel to those of § 3. Conditions that are either necessary or sufficient are easily stated:

Theorem XII. *A necessary condition for a prime ideal is that its elementary-divisor form be a prime function; a sufficient condition is that its norm be a prime function. In other words, the elementary-divisor form of a prime ideal is always a prime function, and the norm of a proper primary ideal is always a proper primary function.*

That the elementary-divisor form of a prime ideal is a prime function was already shown in Theorem I. Suppose now that the norm R_q is equal to a prime function. It is to be shown that, under this hypothesis, every proper divisor $\mathfrak{a}$ of $\mathfrak{q}$ has lower highest dimension; then $\mathfrak{q}$ is recognized as a prime ideal by Theorem II. Indeed, by § 1, 6 the first factor of the norm R_a that differs from the corresponding one of R_q is a proper divisor. Since $R_q = P^{(i)}$, the factor $R^{(i)}$ of R_a must therefore be equal to a proper divisor of $P^{(i)}$, hence to unity; $\mathfrak{a}$ has lower highest dimension.

A second simple proof rests on the following lemma, on which the next theorem also rests.

Lemma VII. *The system of residue classes represented by polynomials $H^{(i)}$ modulo the elementary-divisor form $E^{(i)}$ of a primary ideal – more generally, of an ideal that has only associated prime ideals of dimension $n - i$ – is isomorphic to a subsystem of the residue classes modulo this ideal. The degree of the elementary-divisor form gives the degree of this residue-class system with respect to the quotient field $(P(u, x_{i+1}, \dots, x_n))$.*

Indeed, one defines a correspondence by assigning to one another the classes represented by the same polynomials $H^{(i)}$; sums and products correspond to sums and products. The assignment is also one-to-one. For all polynomials $H^{(i)}$ belonging to the zero class modulo the ideal are, by definition, divisible by $E^{(i)}$; hence the zero class corresponds to the zero class modulo $E^{(i)}$, and conversely as well.

Now let the norm of an ideal be a prime function $P^{(i)}$, hence identical with the elementary-divisor form. The degrees of the residue-class systems modulo the elementary-divisor form $P^{(i)}$ and modulo $\mathfrak{q}$ therefore agree with respect to $(P(u, x_{i+1}, \dots, x_n))$; the subsystem isomorphic to the residue-class system modulo $P^{(i)}$

⁴Likewise the formulation of multiplicity mentioned in note 2 of H.-N. is an immediate consequence of Theorem X. For, after adjoining $x_{i+1}, \dots, x_n$, the residue-class system $\mathfrak{g}_{i-1} \mid \mathfrak{r}_\lambda$ is isomorphic to $\mathfrak{t}_\lambda \mid \mathfrak{r}_\lambda$, where $\mathfrak{t}_\lambda = [\mathfrak{g}_{i-1}, \mathfrak{r}_1, \dots, \mathfrak{r}_{\lambda-1}, \mathfrak{r}_{\lambda+1}, \dots, \mathfrak{r}_a]$ is put, as follows from passing back to modules of linear forms (H.-N., Theorem V) in view of the relative primality of the factors $Q_\mu^{(i)}$. Directly it is shown that $\mathfrak{t}_\lambda \mid \mathfrak{r}_\lambda$ is isomorphic to $\mathfrak{t}_\lambda \mid \mathfrak{q}_\lambda$; here $\mathfrak{t}_\lambda$ arises from the complement of $\mathfrak{q}_\lambda$ after adjoining $x_{i+1}, \dots, x_n$.

therefore exhausts the residue classes modulo $\mathfrak{q}$, and since this system can have no zero divisors, $\mathfrak{q}$ is a prime ideal.

In general Theorem XII cannot be sharpened to necessary and sufficient conditions, as will be shown below by examples. This is possible, however, when the coefficient domain P is a perfect field (§ 1, 14):

Theorem XIII. *If the coefficient domain P is a perfect field, then norm and elementary-divisor form of a prime ideal become prime functions and consequently identical; norm and elementary-divisor form of a proper primary ideal become proper primary functions. Thus, over a perfect field, the condition that the elementary-divisor form be a prime function is necessary and sufficient for a prime ideal; in the condition the norm may also stand in place of the elementary-divisor form.*

Taking Theorem XII into account, Theorem XIII will be proved once it is shown that over a perfect field the ideal becomes prime as soon as its elementary-divisor form is a prime function, and that then the norm is equal to this prime function. In Lemma V, § 3, it was shown that, over a perfect field, the elementary-divisor form becomes a prime function of the first kind as soon as it is a prime function at all. Therefore let $(\mathcal{R})$ denote the residue-class field obtained by adjoining (x_i) to $(P(u, x_{i+1}, \dots, x_n))$, that is, $(P(u, x_i, x_{i+1}, \dots, x_n))$ modulo $P^{(i)}$. Then (x_i) is a zero of the prime function of the first kind $(P^{(i)})$, and hence, as for example Lagrange's formula shows, $(y_1), \dots, (y_n)$ are contained in $(\mathcal{R})$. The subsystem of the residue classes modulo the ideal $\mathfrak{q}$ under consideration that is isomorphic, by Lemma VII, to $(\mathcal{R})$ – where only denominators from $(P(u, x_{i+1}, \dots, x_n))$ occur – therefore contains the residue classes $(y_1), \dots, (y_n)$ and hence exhausts the residue-class system modulo $\mathfrak{q}$. Since, being isomorphic to $(\mathcal{R})$, it can have no zero divisors, $\mathfrak{q}$ is a prime ideal. The degree of this residue-class field $(\mathcal{R})$ of $\mathfrak{q}$ with respect to $(P(u, x_{i+1}, \dots, x_n))$ is equal to the degree of the norm of $\mathfrak{q}$; on the other hand, by the isomorphism with $(\mathcal{R})$, it is equal to the degree of the elementary-divisor form $P^{(i)}$. Thus norm and elementary-divisor form must agree.

For imperfect coefficient domains Theorem XII still allows the following sharpening:

Theorem XIV. *If the coefficient domain P is an imperfect field of characteristic p , then the norm of a prime ideal is the p^g -th power of the elementary-divisor form; one has $0 \leq g \leq (i-1)f$, where f denotes the exponent of the elementary-divisor form and $n-i$ the dimension of the prime ideal.*

Theorem XIV amounts to the assertion that the residue-class field $(\mathcal{R})$ of $\mathfrak{p}$ has degree p^g ($g \geq 0$) with respect to the subfield $(\mathcal{R}') = (P(u, x_i, x_{i+1}, \dots, x_n))$ isomorphic to $(\mathcal{R})$. This follows directly from the Steinitz theorems stated in § 1, 15, more precisely from

Lemma VIII. *Let Λ be a finite extension of the imperfect field Ω of reduced degree r and exponent f ; let Λ_0 be the field of the first kind contained in Λ , and let M be an intermediate field between Λ_0 and Λ . Then the degree of every element of Λ with respect to M is a power of p , where p denotes the characteristic of Ω .*

For the $p^{f'}$ -th power ($f' \leq f$) of every element z of Λ belongs to Λ_0 ; hence z is a zero of a prime function $G(t) = (t - z)^{p^{f'}}$ with coefficients in Λ_0 . Let $H(t) = (t - z)^\delta$ be the prime function in M with zero z . Then $H(t)$ also has coefficients in $\Lambda_0(z)$, and therefore the coefficients of $H(t)$ belong to the intersection field of M and $\Lambda_0(z)$, hence to an intermediate field between Λ_0 and $\Lambda_0(z)$. The degree of z with respect to M is therefore equal to the degree with respect to this intermediate field, and hence, as a divisor of $p^{f'}$, is a power of p .

For the proof of Theorem XIV it remains only to observe that the field $(\mathcal{R}) = (P(u, x_i, x_{i+1}, \dots, x_n))$ must have degree $r \cdot p^f$ with respect to $(P(u, x_{i+1}, \dots, x_n))$, if r denotes the reduced degree and f the exponent of $(\mathcal{R})$; and consequently the field of the first kind $(\mathcal{R}_0)$ contained in $(\mathcal{R})$ must be a subfield of $(\mathcal{R}')$. Since there are elements of degree $r \cdot p^f$ in $(\mathcal{R})$ (§ 1, 15), (x_i) must necessarily be of this degree, since all elements of $(\mathcal{R})$ arise by specializing the $t_{\mu\nu}$ in (x_i) . The exponent of $(\mathcal{R})$ agrees with the exponent of the elementary-divisor form, and $(x_i)^{p^f}$ has degree r and is an element of $(\mathcal{R}_0)$, hence a primitive element of $(\mathcal{R}_0)$. Since $(\mathcal{R})$ arises by adjoining finitely many elements – say $(x_1), \dots, (x_{i-1})$ – to $(\mathcal{R}')$, repeated finite application of Lemma VIII gives the desired proof and at the same time the estimate for g .⁵

We now add the examples which show that Theorem XII cannot in general be sharpened beyond Theorem XIV. They concern a prime ideal whose norm is equal to the square of the elementary-divisor form ($g > 0$), and proper primary ideals whose elementary-divisor forms are prime functions. The latter example also shows that no analogue of Theorem XIV exists here, but that the norm can become an arbitrary power of the elementary-divisor form.

Let the coefficient domain P arise by adjoining two indeterminates λ, μ to the field of residue classes

⁵Compare the parallel considerations in A. Ostrowski, Zur arithmetischen Theorie der algebraischen Größen, Gött. Nachr. 1919, pp. 279–298, especially p. 288, where the corresponding theorem is stated without proof.

modulo 2, and therefore let P be an imperfect field of characteristic two. Consider the ideal

$$\bar{\mathfrak{p}} = (y_1^2 + \lambda, y_2^2 + \mu),$$

which must be a prime ideal, since the residue-class system $(\mathcal{R})$ is isomorphic to the field $P(\sqrt{\lambda}, \sqrt{\mu})$. The field $(\mathcal{R})$ has degree four with respect to (P) , reduced degree $r = 1$, and exponent $f = 2$. Therefore the elementary-divisor form $P^{(2)}$ is of the second degree, and the norm is of the fourth degree and consequently the square of $P^{(2)}$, as can also be checked directly from the module representation (3), § 1, 5. One obtains

$$P^{(2)} = (u_{11}u_{22} + u_{12}u_{21})^2 x_2^2 + (u_{11}^2 \mu + u_{21}^2 \lambda),$$

or, after a suitable multiplication,

$$x_2^2 + (t_{12}^2 \lambda + t_{22}^2 \mu).$$

Let P now arise by adjoining the one indeterminate λ to the field of residue classes modulo 2, and take as basis the ideal

$$\bar{\mathfrak{q}} = (y_1^2 + \lambda, y_2^2 + \lambda) = (y_1^2 + \lambda, (y_1 + y_2)^2).$$

As elementary-divisor form, by specializing $\mu = \lambda$ in the one just given, one obtains

$$P^{(2)} = x_2^2 + \lambda(t_{12}^2 + t_{22}^2),$$

hence a prime function, since $t_{12} = (0)$, $t_{22} = (1)$ leads to the prime function $x_2^2 + \lambda$. But $\bar{\mathfrak{q}}$ is proper primary, since it contains $(y_1 + y_2)^2$ but not $(y_1 + y_2)$. The norm must be equal to the square, hence to the p -th power, of $P^{(2)}$, since the number of residue classes of $\bar{\mathfrak{q}}$ linearly independent over P is four.

That this last analogue of Theorem XIV is not always fulfilled is shown by the following example. Let P arise by adjoining the indeterminate λ to the residue-class field modulo 3, and put

$$\mathfrak{q} = (y_1^3 + \lambda, (y_1 - y_2)^2).$$

Then, as above, $\mathfrak{q}$ is a proper primary ideal. The elementary-divisor form becomes – taking into account that $y_2^3 + \lambda$ is also divisible by $\mathfrak{q}$ –

$$P^{(2)} = x_2^3 + \lambda(t_{12} + t_{22})^3,$$

a prime function; but the norm is equal to the square of $P^{(2)}$, since there are six linearly independent residue classes modulo $\mathfrak{q}$. Thus the p^g -th power of the elementary-divisor form does not occur here.

The first example shows that, over an imperfect field as coefficient domain, just as in algebraic number fields, prime ideals of degree higher than the first may occur, where p^g is to be called the degree of the prime ideal. The further examples are to be understood as analogues of ramification ideals: a prime number may be divisible by a higher power of a prime ideal, by a primary ideal. For, as shown in note 10, the elementary-divisor form corresponds to the least integral rational number divisible by an ideal in an algebraic number field.

§ 7. Absolute Prime Ideals

A prime ideal with coefficients in P is called an *absolute prime ideal* if it remains a prime ideal in the algebraically closed field A to which P can be extended.⁶ Correspondingly, a prime function P with coefficients in P is called an *absolute prime function* – absolutely irreducible polynomial – if P remains a prime function in A . The characterization of absolute prime ideals is given by

Theorem XV. *A necessary and sufficient condition for an absolute prime ideal is that its elementary-divisor form – which may be assumed integral and primitive in the u – become an absolute prime function as a polynomial in the x and u . The elementary-divisor form becomes identical with the norm, so that the condition may also be stated for the norm.*

For the proof, observe first that in general norm and elementary-divisor form are preserved under algebraic extension of the coefficient domain. Indeed the individual factors $R^{(i)}$ respectively $E^{(i)}$ are defined as greatest common divisors in the polynomial sense, a property preserved by algebraic extension of the coefficient domain. Thus $P^{(i)}$ remains the elementary-divisor form of $\mathfrak{p}$ when passing from P to A , hence when passing to a perfect field as coefficient domain, since every algebraically closed field is perfect, as the definition

⁶An algebraically closed field is, as is well known, a field in which every polynomial in one indeterminate decomposes into linear factors. For the existence and essential uniqueness of the algebraically closed field belonging to an arbitrary field, see Steinitz.

immediately shows. By Theorem XIII, a necessary and sufficient condition for $\mathfrak{p}$ to be an absolute prime ideal is that $P^{(i)}$ become a prime function with respect to $A(u)$; that is, an absolute prime function as a polynomial in the x and u , if $P^{(i)}$ is assumed integral and primitive in the u . At the same time, by Theorem XIII, $P^{(i)}$ becomes identical with the norm of $\mathfrak{p}$ as soon as A is taken as coefficient domain. Hence, by what was just noted, the same holds over P . This proves Theorem XV.

For absolute prime functions P with coefficients in a (finite) algebraic number field $\mathfrak{K}$, one now has the theorem⁷ that P remains a prime function, more precisely an absolute prime function, modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted. The transfer of this theorem to absolute prime ideals rests on

Theorem XVI. *If, as coefficient domain, one takes, in place of a field, the ring $\mathfrak{o}^*$ of all algebraic integers of a (finite) algebraic number field $\mathfrak{K}$, then norm and elementary-divisor form of a polynomial ideal are preserved as norm and elementary-divisor form modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted.*

For the proof the polynomial domain on which everything is based is to be modified relative to § 1, 1. Let $\tilde{\mathfrak{S}}$ consist of all polynomials in $y_1, \dots, y_n$ with coefficients in $\mathfrak{o}^*$, hence algebraic integers of $\mathfrak{K}$; let the coefficient domain of $\mathfrak{S}$ be $\mathfrak{o}^*[u]$, that is, all polynomials in u with coefficients in $\mathfrak{o}^*$. If $\bar{\mathfrak{m}}$ is an ideal in $\tilde{\mathfrak{S}}$, it passes by (1) into a transformed ideal $\mathfrak{m}$ in $\mathfrak{S}$. It is to be shown that only finitely many polynomials from $\mathfrak{o}^*[u]$ occur in the denominators in forming the norm; then Theorem XVI follows directly.

First note that the module representation (3) in § 1, 5, for the formation of the individual norm, amounts to reducing powers of x_{i-1} modulo a polynomial regular in x_{i-1} ,

$$C^{(i-1)} = U_{i-1}(u)x_{i-1}^{e_{i-1}} + \text{lower terms},$$

⁸ where all coefficients of $C^{(i-1)}$, in particular U_{i-1} , may be assumed to be elements of $\mathfrak{o}^*[u]$. Since this is a matter of successive reduction of $x_1, \dots, x_{i-1}$, the module representation integral in the x is also integral in $\mathfrak{o}^*[u]$ except for power products

$$U_1^{\lambda_1} \dots U_{i-1}^{\lambda_{i-1}}$$

in the denominator. The product $U_1 U_2 \dots U_n$, considered as a polynomial in the u , defines by its coefficients an ideal $\mathfrak{r}^*$ of $\mathfrak{K}$, which is therefore divisible by only finitely many prime ideals of $\mathfrak{K}$. If $\mathfrak{p}^*$ is chosen different from these finitely many, and if the residue-class field modulo $\mathfrak{p}^*$ is used as coefficient domain, the original module representation is preserved by replacing each number of $\mathfrak{o}^*$ by its residue class modulo $\mathfrak{p}^*$.

Furthermore, since norm and elementary-divisor form are determined only up to factors from $\mathfrak{o}^*[u]$, they may also be assumed divisible by $\mathfrak{m}$ with respect to $\mathfrak{o}^*[u]$. Suppose, under this convention, that for instance

$$R^{(i)} = V_i(u)x_i^{e_i} + \text{lower terms},$$

so that, in the divisibility of all ρ -rowed determinants of the module $\mathfrak{M}_{i-1}^*$ of rank ρ determined by $C^{(i-1)}$, only powers of V_i occur in the denominator besides the $U_1^{\lambda_1} \dots U_{i-1}^{\lambda_{i-1}}$. Choose $\mathfrak{p}^*$ further distinct from the finitely many prime ideals of $\mathfrak{K}$ that occur in the ideal determined by $V_1 V_2 \dots V_n$. Then $R^{(i)}$ remains a common divisor of these determinants when the residue-class field modulo $\mathfrak{p}^*$ is taken as coefficient domain, and the rank of $\mathfrak{M}_{i-1}^*$ is preserved, since $C^{(i)}$ was a ρ -rowed determinant from $\mathfrak{M}_{i-1}^*$.

Finally $\mathfrak{p}^*$ is to be chosen so that $R^{(i)}$ remains the greatest common divisor in each case. Let $D^{(i)}$ run through all ρ -rowed determinants from $\mathfrak{M}_{i-1}^*$; by hypothesis

$$U_1^{\lambda_1} \dots U_{i-1}^{\lambda_{i-1}} V_i^\rho D^{(i)} = R^{(i)} T^{(i)}$$

with the $T^{(i)}$ having no common divisor containing x in the polynomial sense. Consequently the ideal derived from all $T^{(i)}$ contains a polynomial

$$G^{(i+1)} = W_i(u)x_{i+1}^{m_i} + \text{lower terms}, \quad W_i \neq 0.$$

That $G^{(i+1)}$ may be assumed regular in x_{i+1} follows from composing the transformation (1) with one of $x_{i+1}, \dots, x_n$ having new indeterminates as coefficients. If, then, $\mathfrak{p}^*$ is also chosen distinct from the finitely many prime ideals contained in the ideal determined by $W_1 W_2 \dots W_n$, then when the residue-class field

⁷A. Ostrowski, loc. cit. (note 21), lemma p. 296. A simpler proof is in E. Noether, Ein algebraisches Kriterium für absolute Irreduzibilität, Math. Ann. 85 (1922), pp. 26–33, no. 7.

⁸Compare H.-N., § 4.

modulo $\mathfrak{p}^*$ is used as coefficient domain, $R_{\mathfrak{m}}$ is in fact preserved as norm. Quite correspondingly $\mathfrak{p}^*$ can be chosen still further so that $E_{\mathfrak{m}}$ also remains the elementary-divisor form. This proves Theorem XVI.

From Theorems XV and XVI, as a generalization of the theorem on absolute prime functions and using that theorem, one obtains

Theorem XVII. *If $\mathfrak{p}$ is an absolute prime ideal with algebraic integers from a (finite) algebraic number field $\mathfrak{K}$ as coefficients, then $\mathfrak{p}$ remains a prime ideal, more precisely an absolute prime ideal, modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted.*

For the proof, let the norm $R_{\mathfrak{p}}$ of $\mathfrak{p}$ be chosen, as in Theorem XVI, so that it is divisible by $\mathfrak{p}$ also with respect to $\mathfrak{o}^*[u]$. Then

$$R_{\mathfrak{p}} = T(u)P^{(i)}(x),$$

where $P^{(i)}(x)$ may be assumed integral and primitive in the u . By Theorem XV, $P^{(i)}$, regarded as a polynomial in u and x with coefficients in $\mathfrak{K}$, is an absolute prime function. By the theorem on absolute prime functions it retains this property modulo every prime ideal of $\mathfrak{K}$, with at most finitely many prime ideals of $\mathfrak{K}$ excepted. Thus choose $\mathfrak{p}^*$ different from these finitely many prime ideals and, by Theorem XVI, also different from finitely many further prime ideals of $\mathfrak{K}$. Then $P^{(i)}$ – its coefficients from $\mathfrak{o}^*$ being replaced by the corresponding residue classes modulo $\mathfrak{p}^*$ – becomes the norm of $\mathfrak{p}$ when the residue-class field modulo $\mathfrak{p}^*$ is taken as coefficient domain P , and this norm is an absolute prime function. Taking P as coefficient domain, $\mathfrak{p}$ therefore remains an absolute prime ideal by Theorem XV. This proves Theorem XVII.

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